Abstract algebra – 4

Group actions, homomorphisms and automorphisms

Homework

13 сентября 2026 года

Theoretical minimum. An action of G on X is a homomorphism $G \to \mathrm{Sym}(X)$; the orbit of x has $[G : G_x]$ elements, where G_x is the stabiliser, so every orbit size divides $|G|$. The number of orbits equals $\frac{1}{|G|}\sum_g |\operatorname{Fix}(g)|$ (Burnside), which is proved by counting the pairs (g, x) with $gx = x$ in two ways. Left translation gives an embedding $G \hookrightarrow S_{|G|}$ (Cayley); the action on the cosets of a subgroup H of index n gives a homomorphism $G \to S_n$ with kernel the largest normal subgroup contained in H, whence a subgroup of smallest prime index is normal. Conjugation gives $\mathrm{Inn}(G) \cong G/\operatorname{Z}(G) \trianglelefteq \mathrm{Aut}(G)$. Finally $\mathrm{Aut}(\mathbb{Z}/n) \cong (\mathbb{Z}/n)^*$ and $\mathrm{Aut}((\mathbb{Z}/p)^k) \cong \mathrm{GL}_k(\mathbb{F}_p)$.

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1. Let G be a finite group. For subsets $U, V, W \subseteq G$ denote by N_{UVW} the number of triples $(x, y, z) \in U \times V \times W$ with $xyz = e$. Suppose G is partitioned into three sets A, B, C. Prove that $N_{ABC} = N_{CBA}$.
(IMC)

2. Suppose that a finite group G is generated by two elements g and h, where the order of g is odd. Show that every element of G can be written in the form $g^{m_1}h^{n_1}g^{m_2}h^{n_2}\cdots g^{m_r}h^{n_r}$ with $1 \leqslant r \leqslant |G|$ and all exponents in $\{-1, 1\}$. (The IMC problem on the non-existence of an injective homomorphism $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z}) \to S_p$ is Linear algebra – 9, homework 6.) (Putnam 2016, A5)

3. Let G be a finite group with $n \geqslant 2$ elements and let p be the largest prime divisor of n. Prove that the number of endomorphisms $f\colon G \to G$ is at most $n^{n/p}$, and find all groups for which equality holds.
(Румыния, 2008)

4. For a group G and $m \in \mathbb{N}$ let $G(m)$ be the subgroup generated by

all m-th powers of elements of G. Prove that if $G(m)$ and $G(n)$ are commutative, then so is $G(\gcd(m,n))$. (IMC)

5. Prove that $\mathrm{Aut}(\mathbb{Z}/n) \cong (\mathbb{Z}/n)^*$ and that $\mathrm{Aut}((\mathbb{Z}/p)^k) \cong \mathrm{GL}_k(\mathbb{F}_p)$, and compute the order of the latter. Then find a finite group G with $|\mathrm{Aut}(G)| = 1$ and prove that there are exactly two such groups.

6. Using Burnside's lemma, count the colourings of the faces of a cube in k colours up to rotations of the cube, and the colourings of its vertices; then explain why the two answers are polynomials in k with rational but non-integral coefficients whose values are nevertheless integers.

7. Let G be a group whose operation is $*$ such that (i) G is a subset of $\mathbb{R}^3$, the operation being unrelated to vector addition, and (ii) for all $a, b \in G$ either $a \times b = a * b$ or $a \times b = 0$, where $\times$ is the usual cross product. Prove that $a \times b = 0$ for all $a, b \in G$. (Putnam 2010, A5)

8. Let a finite group G act transitively on a set X with $|X| > 1$. Prove that some $g \in G$ has no fixed point at all (Jordan), and show that the proportion of such g is at least $1/|X|$.

9. Prove that $\mathrm{Inn}(G)$ is normal in $\mathrm{Aut}(G)$, and that every automorphism of S_n which maps transpositions to transpositions is inner. (For $n \neq 6$ every automorphism does map transpositions to transpositions, so $\mathrm{Aut}(S_n) = \mathrm{Inn}(S_n)$; the case $n = 6$ is a genuine exception.)

10. Prove that a group with a subgroup of index n has a normal subgroup of index dividing $n!$, and deduce that a simple group of order 60 has no subgroup of index less than 5.